Evaluating the Association Between Polycyclic Aromatic Hydrocarbon Exposure, Epigenetic Age Acceleration, and Lung Cancer Risk

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Abstract

Polycyclic aromatic hydrocarbons (PAHs) are environmental carcinogens linked to DNA methylation changes and lung cancer, yet whether PAH exposure accelerates epigenetic aging, and whether this contributes to lung cancer risk, has not been systematically evaluated. This review searched PubMed, Google Scholar, and ScienceDirect, identifying 53 potentially relevant records. Following removal of duplicates and title/abstract and full-text screening through PRISMA guidelines, 33 studies met inclusion criteria. Each study was assessed by one of five reviewers and extracted into a standardized evidence matrix capturing study design, population, exposure biomarkers, methylation methodology, epigenetic clock, effect estimates, and evidence classification (primary, mechanistic, or background). PAH exposure was consistently associated with altered nuclear and mitochondrial DNA methylation, oxidative stress, and increased lung cancer risk. Evidence linking PAH exposure to epigenetic age acceleration was moderate to strong, particularly using second-generation clocks, while the association between epigenetic aging and lung cancer risk was supported observationally but not confirmed by Mendelian randomization. No single study evaluated the complete pathway of PAH exposure to epigenetic aging to lung cancer within one population. This review reconstructs this pathway from complementary evidence, identifies epigenetic aging as an appropriate biomarker rather than a proven causal driver, and highlights the necessity for prospective cohorts measuring exposure, epigenetic clocks, and lung cancer outcomes in the same population.

I. Introduction

Polycyclic aromatic hydrocarbons (PAH) are a class of organic compounds composed of two or more carbon rings covalently bonded with peripheral hydrogen. They originate from incomplete combustion through pyrolysis of organic material where carbon doesn't oxidize due to insufficient oxygen. PAHs can be divided into low-molecular-weight (2 to 3 rings; naphthalene, fluorene, phenanthrene) and high-molecular-weight types (4 to 5 rings; benzo[a]pyrene (B[a]P), benz[a]anthracene), with the inclusion of methyl types (added methyl group; 5-methylchrysene). These are classified by the International Agency for Research on Cancer based on the amount of carcinogenic evidence gathered. (Alhamdow et al., 2020). Exposure to PAHs is inherently a mixture exposure, which

leads to studies relying on summary measurements (∑OH-PAHs) and mixture statistics rather than single-compound measurements.

Humans are exposed to PAHs through various types of sources. Studies based on occupational exposure are commonly represented throughout our information: coke-oven work (Campisi et al., 2023) and chimney sweeping and creosote impregnation (Alhamdow et al,. 2018). Household PAH exposure can result from indoor combustion of solid fuels for cooking and heating. (Blechter et al., 2023). Environmental factors can include residential proximity to industrial areas and community exposure in urban districts (Zhou et al., 2026). Finally, individual sources of PAHs exposure include tobacco smoke, in active forms (Khan et al., 2026) and secondhand forms (Patil et al., 2025), along with charred foods and high-temperature cooking. Routes of exposure to PAHs involve inhalation, ingestion, and dermal absorption; in a design from one of our revised studies (Alhamdow et al., 2020).

The classic pathway from PAH exposure to cancer is metabolic activation. Xenobiotic metabolizing enzymes attempt elimination through conversion of PAHs into reactive intermediates. B[a]P binds the aryl hydrocarbon receptor (AhR), upregulating cytochrome P450 enzymes CYP1A1 and CYP1B1 (Liu et al., 2026; Alhamdow et al., 2018). These enzymes, along with epoxide hydrolase, function to convert B[a]P into benzo[a]pyrene diol epoxide (BPDE), a reactive intermediate that covalently binds to DNA to form mutagenic adducts that can cause mutations, including in cancer-related genes such as TP53. These molecular mechanisms are consistent with human evidence linking PAH exposure to lung-cancer risk.

DNA methylation is the addition of a methyl group to a cytosine base at a CpG site, where cytosine sits adjacent to a guanine base in a single strand of DNA. As an epigenetic modification, it alters gene transcription without changes in the underlying sequence. Its function depends on the genomic context: promoter methylation typically suppresses transcription, although methylation does not universally reduce gene expression. Measurement approaches in this literature range from targeted single-locus assays, methylation-specific PCR (Liu et al., 2026) and bisulfite pyrosequencing (Alhamdow et al., 2018); to genome-wide arrays (Meng et al., 2021; Chen et al., 2024). Related epigenetic marks examined in this literature include mitochondrial DNA (mtDNA) methylation (Guo et al., 2022), and microRNA alteration (Xiao et al., 2024).

DNA methylation patterns across CpG sites underlie measurements of epigenetic aging, a proxy for biological aging; the progressive decline in cellular and tissue function that occurs at varying rates between individuals of similar chronological age. Methylation at certain CpG sites predictably changes with chronological age, where epigenetic clock biomarkers are weighted formulas over a set of these sites estimating epigenetic age. Epigenetic age acceleration is the residual obtained by regressing an individual's clock-estimated epigenetic age based on their chronological age, such that positive values indicate an individual is biologically older than expected for their age. On one hand, First-generation clocks, such as Horvath and Hannum, are trained to predict chronological age (Blechter et al., 2023; Shi et al., 2022; Zhang et al., 2025). Second-generation clocks, on the other hand, have PhenoAge and GrimAge, which are trained on mortality and clinical outcomes rather than calendar age (Li et al., 2022; Rahman et al., 2024; Cardenas et al., 2022). Because aging is a major risk factor for most cancers, biological aging compresses the timeline over which modifications and alterations in cancer accumulate.

Each segment of this research’s pathway has been supported to varying degrees in separate bodies of literature. Studies connecting PAH exposure to epigenetic aging, epigenetic aging to lung cancer, and PAH exposure to lung cancer have been conducted through different populations,

biomarkers, samples, and other factors; their differences show difficulty in integrating evidence and determining the chance that the completed pathway occurs in the same individuals. This systematic review therefore evaluates published evidence across PAH exposure to the DNA methylation/epigenetic aging pathway to aim to evaluate the question concerning the correlation of PAH exposure to epigenetic age acceleration, DNA methylation, and lung cancer.

II. Methodology

a. Overview and Aims

This systematic literature review aims to investigate the association between environmental exposure to Polycyclic Aromatic Hydrocarbons (PAHs) and epigenetic age acceleration in human lung cancer by reconstructing and synthesizing findings from multiple studies, as no single study fully captures the pathway. This paper aims to focus on the following direct evidence pathways: (1) PAH exposure to DNA methylation; (2) PAH exposure to epigenetic aging; (3) Epigenetic aging to lung cancer; (4) PAH exposure to lung cancer. In addition to these direct pathways, this paper utilizes supporting mechanistic evidence to solidify the pathway's progression. Mechanisms covered within the evidence base include PAH metabolic activation, reactive metabolites, oxidative stress, DNA damage, BPDE-DNA adduct formation, cell cycle changes, and epigenetic mechanisms.

This systematic review is conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (utilizing the PRISMA 2020 flow diagram for new systematic reviews, which included searches of databases and registers only). This was achieved utilizing a record-level screening tracker for identification, screening, and inclusion of studies (presented within a flow diagram), followed by an evidence matrix in order to compile data items of included studies, and finally, synthesis, evidence mapping, and heterogeneity analysis of studies in groupings of pathways.

b. Search Procedures

To begin, a literature search was performed through the databases of PubMed, ScienceDirect, and Google Scholar for relevant records. Google Scholar was last searched on 8/3/26. PubMed was last searched on 8/2/26. ScienceDirect was last searched on 7/28/26. The following search terms were utilized during the screening with boolean OR and AND operators: (1) "PAH exposure"; (2) "DNA methylation"; (3) "Lung cancer"; (4) "Epigenetic age acceleration"; (5) "Epigenetic aging"; (6) "benzo[a]pyrene"; (7) "DNA methylation change"; (8) "Cancer-related damage"; (9) "Human"; (10) "Epigenetic clock"; (11) "Methylation clock"; (12) "Epigenetic age"; (13) "mtDNA methylation"; (14) "Epigenetics"; (15) "Lung cancer study": (16) "Polycyclic aromatic hydrocarbons"; (17) "Workers"; (18) "Mechanism"; (19) "Lung tissue"; (20) 'Carcinogenesis"; (21) "Biological sample"; (22) "Cross sectional". Filters were used to limit studies to a particular time frame, study type, and text availability status. These varied by pathway and search objective.

c. Record-level screening

Results that were considered potentially relevant were identified and put into a record-level screening tracker. The tracker included details of database, search date, exact search string, filters used, number of results, full citation, publication year, DOI/PMID, and full text accessible link. Once inserted into the tracker, the record was screened in context of the inclusion and exclusion criteria by 1 of 5 reviewers (SA, NA, EK, GK, SM) with the following order: (1) Identification of duplicate studies; (2) Title and abstract screening; (3) Full text retrieval; (4) Full text eligibility screening; (5) Full text inclusion/exclusion decision; (6) Assignment to evidence pathway [mentioned within 3.1]; (7) Assignment to evidence category (primary evidence, mechanistic evidence, supporting/background evidence.

In the identification of the eligibility of studies in the screening process, the following inclusion criteria was utilized: (1) Is original research; (2) Is published in a peer-reviewed journal; (3) Measures PAH exposures or related biomarkers; (4) Examines an epigenetic/DNA methylation/aging/cancer related outcome; (5) Fits at least one examined evidence pathway [mentioned in 3.1]; (6) Provides usable methods and results; (7) Has full-text availability; (8) Is written in English; (9) Is a human observational study or animal/cell mechanistic study.

Similarly, studies were excluded from our review if they met the following criteria: (1) Discusses general air pollution rather than separating specific PAH exposures; (2) Discusses outcomes or pathways other than epigenetics or cancer; (3) Fits none of the examined evidence pathways; (4) Is a review; (5) Is an editorial, news article, blog, or other educational source [not in a scientific journal]; (6) Is a duplicate publication already mentioned within our review; (7) Is lacking methods or results for us to draw a proper conclusion; (8) Lacks full-text availability.

Once studies were screened by 1 of 5 reviewers, a check was run over the entire entry of a record into the record-level tracker by 1 of 2 mentors (SG, VM). A final inclusion/exclusion decision as well as assignment to evidence category was confirmed by the mentors. Any uncertain studies that reviewers identified were given an inclusion/exclusion reason by mentors.

Evidence categories were used to determine the future extent of a record's inclusion within the review. Studies marked as excluded by mentors were considered outside the scope of the review and therefore excluded from the review. Studies marked as background/support evidence only were considered as only partially relevant and therefore excluded from the review but utilized for the understanding of the reviewers. Studies marked as mechanistic (primarily but not limited to cell and animal models) were focused on mechanisms of carcinogenesis and epigenetic change rather than direct outcomes on a population, which were utilized within the final pathway reconstruction and therefore included within the review. Lastly, studies marked as primary were human observational/clinical evidence supporting pathway outcomes/biomarkers within a population and were included as direct evidence within the review.

Within the tracker, a total of 53 potentially relevant records were identified. After screening, 33 studies were determined as suitable to include within the literature review. The full process, including identification, screening and inclusion, is visually represented in Figure 1 [PRISMA flowchart].

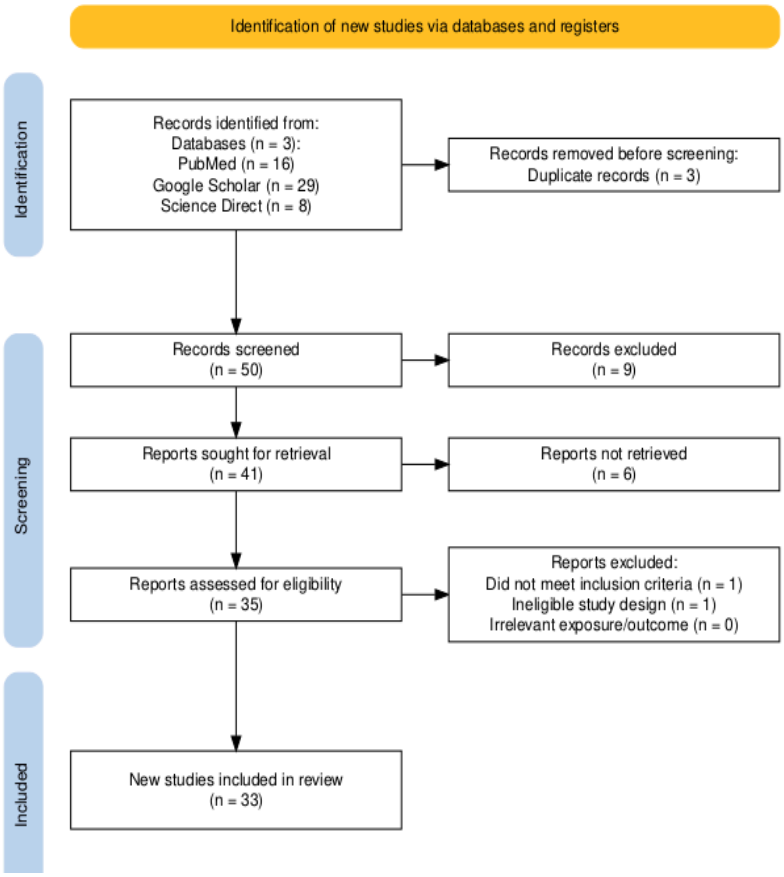


Figure 1: PRISMA flow chart of systematic review for PAH exposure, epigenetic age acceleration and lung cancer

d. Data extraction

The 33 relevant studies were then transferred to an evidence matrix for the detailed extraction of data in a format that is easily accessible for synthesis. The full text of each study was assessed by 1 of 5 reviewers to determine the study design and statistical findings.

To determine study design and preliminary details, the following subjects were identified and stored within the matrix: full citation, DOI/PMID, full text link, publication year, study objective, study design, human/animal/cell classification, populational/experimental model, participant characteristics, sample size, geographic location, biological sample or tissue, PAH exposure source, exposure measurement or biomarker, epigenetic outcome, methylation assay or measurement method, DNA methylation clock, lung cancer relevance. Then, to determine statistical outcomes and final study interpretation, the following subjects were identified and stored within the matrix: main finding, effect direction, effect estimate, confidence interval, p-value, potential confounding variables, variables adjusted for, proposed mechanism, author's conclusion, cohort's interpretation, strengths and limitations, evidence pathway, and finally, the inclusion or exclusion decision.

Any areas not present within a study were marked as NR, and areas not applicable/appropriate to the design of a study were marked as NA. Uncertainties by reviewers about data were marked and then addressed by mentors to ensure a full and accurate evidence base for synthesis.

e. Data Analysis

The data were then analyzed for patterns within evidence pathways. Each reviewer was assigned to the following respective evidence pathways: (1) PAH exposure to DNA methylation alterations to lung-cancer or lung-related outcomes; (2) PAH exposure to epigenetic age acceleration; (3) Epigenetic or biological aging to lung-cancer risk; (4) PAH exposure to lung-cancer risk; (5) MECHANISTIC: PAH exposure to oxidative stress/DNA damage to epigenetic change and carcinogenesis.

For each pathway, reviewers identified studies that were relevant to their pathways. In a shared analysis document,3-7 included studies were noted with their citation, study design, population/model, evidence category, and the reason for their inclusion within the pathway synthesis. There were instances of overlap between the pathways depending on study relevance. After all relevant studies for each pathway had been transferred from the evidence matrix to the analysis document, the following uniform synthesis areas were identified: (1) Overall patterns across studies; (20 Studies provide the strongest support for that pattern; (3) Inconsistent, mixed, nonsignificant, or substantially indirect studies; (4) Differences may explain the variation; (5) Overall evidence strength: strong, moderate, limited, inconsistent, mechanistically plausible, or not adequately studied. This information was utilized to create a finalized synthesis statement.

Once synthesis of evidence had been conducted through this mapping structure, each reviewer completed an analysis of heterogeneity within their studies on the following focuses: (1) Exposure source and exposure measurement; (2) DNA methylation clocks and aging measures; (3) Outcome definition and study design; (4) Exposure setting, population, and confounder control; (5) Experimental model, tissue, and mechanistic endpoint. For each heterogeneity focus, the following areas were identified to compile a finalized heterogeneity statement: (1) Most important source of heterogeneity; (2) Effect on the strength of the conclusion; (3) Comparability of studies based on heterogeneity.

These pathways and heterogeneity analysis mentioned above were utilized to develop the results synthesis covered in the remaining sections of this review.

III. Study Characteristics

a. General Characteristics (Humanistic vs Mechanistic Evidence, Study Designs, Population, and PAH Exposure)

For this review, a total of 33 peer-reviewed studies with full-text availability were synthesized. Nearly 73% of the included studies had human populations, as they primarily focused on the association between PAH exposure, epigenetic age acceleration, and lung cancer risk, rather than determining causal mechanisms that prove the association is present across diverse demographics, PAH exposures, sample types, epigenetic/DNAm clocks, and exposure biomarkers. However, the remaining mechanistic evidence aimed to understand possible association-related underlying cellular mechanisms.

Cross-sectional, longitudinal, case-control, repeated-measures, in vitro, and Mendelian randomization designs were mainly used. Each design addresses different research objectives. However, cross-sectional designs predominated (14 of 33), followed by case-control (8) and longitudinal (2). The vast majority of the studies had cross-sectional and case-control designs, which limited temporal and causal inference. The lower frequency of repeated-measures, in vitro, and other designs demonstrated

that evidence was limited on long-term results and mechanistic understanding (Shi et al., 2022; Zhang et al., 2023; Khan et al., 2026).

Populations varied due to sex and age, ranging from early life to adult-to-old-adult exposure, as well as both sexes or female-only focused populations (Patil et al., 2025; Shi et al., 2022; Blechter et al., 2023). Exposure type was also another variable, since studies included occupational, tobacco smoke, household or urban pollution, and second-hand smoke exposure (Patil et al., 2025; Shi et al., 2022; Khan et al., 2026; Olsson et al., 2022; Blechter et al., 2023).

Smoking status was one of the most important factors since tobacco smoke was shown to be a major source of PAH exposure and also to have a role in DNA methylation and lung cancer risk (Zhang et al., 2023; Chen et al., 2024). Studies differed in their inclusion of smokers, former smokers, never-smokers, or mixed populations (in the case of case-control studies).

Geographically, a substantial proportion of research centered on populations in China (Wuhan, Shanghai, Xuanwei, etc), with fewer populations concentrated in Europe and the United States, highlighting differences that may reflect diverse environmental and occupational exposure contexts (Shi et al., 2022; Olsson et al., 2022; Zhang et al., 2025).

b. Specific Characteristics (Biological Samples, PAH Exposure Assessment, DNAm and Epigenetic Clocks)

Other more specific characteristics were taken into account as part of the heterogeneity. Urine samples were commonly used to measure internal PAH exposure through OH-PAH metabolites, while blood samples were used for DNA methylation measurement (Shi et al., 2022; Alhamdow et al., 2019; Li, J et al., 2018). Additionally, samples of buccal and lung cells were observed, which could potentially provide a substantially tissue-specific analysis of lung-related processes (Patil et al., 2025; Khan et al., 2026).

Regarding DNAm clocks to calculate epigenetic age acceleration, two types of DNAm clocks were used in epigenetics-focused studies: first-generation - Horvath, Hannum, Zhang - and second-generation - PhenoAge, GrimAge, etc. (Shi et al., 2022; Blechter et al., 2022; Patil et al., 2025; Xu et al., 2025). On the other hand, exposure assessment primarily relied on a few biomarkers such as BPDE-albumin adducts and urinary PAH metabolites, reflecting different measurements for human exposure and possible PAH exposure-related DNA methylation changes (Zhang et al., 2025; Khan et al., 2026). The heterogeneity between methodologies reduces the direct comparability of effect estimates and limits generalizability across exposure settings and populations.

IV. Results I - PAH exposure and DNA Methylation

Across the included studies, PAH exposure was associated with DNA methylation differences at multiple CpGs (cytosine-guanine sites). Even so, papers differed in the loci analyzed, their changes due to PAH specificity, exposure type, and study population.

a. Nuclear gene-specific methylation (AHRR/F2RL3 + CYP1A1/GST genes)

A particular pattern was found in AHRR genes. For instance, cg05575921, a critical CpG previously associated with tobacco smoking (a main PAH exposure source), showed a linear regression model of (95% CI) = –8.2 (–12.0 to –4.6) in highly exposed groups (chimney sweeps versus control, along with smokers-only model), but fluorene displayed the most significant PAH-specific association with hypomethylation. Whereas phenanthrene and pyrene did not show a clear association. A study conducted a genome-wide sequencing analysis using the Illumina EPIC BeadChip and identified hypomethylation at a different CpG site, cg08238319, in high exposure groups (Chen et al. 2024). Differing from cg05575921, another study focused on a group exposed to a mixture of PAHs and other industrial pollutants (Chen et al., 2024). This finding demonstrates a similar AHRR-related methylation pattern; however, the exact CpG site and exposure context varied among studies. Smoking was a confounding variable that complicated causal interpretations because it is linked to methylation at AHRR and F2RL3 and is a major PAH source. Despite that, the study primarily focused on one PAH, 1-OHP, in a mix of mercury and vanadium, which caused it to potentially not associate these effects solely with PAH exposure, as Alhamdow et al. (2018) report.

On the other hand, F2RL3 results were only demonstrated in the Alhamdow et al. (2018) study due to its chimney sweep population sample (specific high exposure), with F2RL3 previously associated with lung cancer risk in other studies mentioned by these authors. The F2RL3 probes, similar to AHRR-specific probes previously discussed, experienced hypomethylation, as it provided F2RL3_CpG1 B = −4.75 and F2RL3 cg03636183 B = −4.93 in chimney sweeps. Although a different gene, the same study shows higher hypomethylation in smoker chimney sweeps than in non-smoking workers, as the latter only showed a low hypomethylation (F2RL3 average B = - 0.81), which is a conclusion reached in the same study and in Chen et al. (2024) with AHRR genes due to a specific portion of populations in both studies exposed to smoking.

When it comes to CYP1A1 and GST genes, on the other hand, another comparison is made. Higher urinary 1-OHP exposure was associated with the altered methylation of CYP1A1, GSTP1, and GSTM1; however, the direction and magnitude varied between genes: CYP1A1 methylation B 0.13, GSTM1 methylation B 0.14, GSTP1 methylation B 0.15; CYP1A1 mRNA B −0.40, GSTP1 mRNA B −0.16, GSTM1 mRNA B −0.04. This is due to the low percentage change in DNA methylation presented in high-exposure groups: CYP1A1 4.07; GSTP1 6.02; GSTM1 8.23; CYP1A1 mRNA 0.20; GSTP1 0.55 (Liu et al., 2026). These findings do not support a uniform directional pattern, but rather suggest that gene-specific methylation responses occur with PAH exposure. Results also show that both hypo- and hypermethylation were present. However, only one study was able to demonstrate the results of both genes, showing that evidence is limited in terms of replication.

b. Mitochondrial methylation (mt-atp6/8 + other mtDNA loci)

Both of the Guo et al. (2022) studies showed heterogeneous mitochondrial methylation response across exposure settings and loci, but the mtDNA probes reported similar results in both studies. MT-CO1-P2 β=1.66 and tRNA-Leu-P2 β=1.64 were detected in Guo et al. (2022a), with a population that included with and without lung cancer individuals, while (with a different type of results set) in Guo et al (2022b), had a DNAm percentage decrease change of 5.29 for MT-ATP6 and 7.58 for MT-ATP8,

for a population that only included healthy adults, with both studies using the same methylation assay (Bisulfite PCR pyrosequencing), although the latter also used platelet mtDNA. Guo et al. (2022a) also showed that oxidative stress was present when the population had environmental PAH exposure, which Guo et al. (2022b) did not report, yet both reported their mtDNA loci hypomethylation with increasing PAH/PM2.5 exposure, despite the populations differing. These differing findings indicate that PAH-associated mtDNA methylation is exposure- and locus-specific instead of uniformly hypermethylated or hypomethylated.

c. Genome-wide and other CpG findings

Genes such as ACBD6, SAMD4A, and UBE2O were also reported to have changes, especially a CpG annotated to ACBD6 that was linked to lung-cancer risk in the study. It showed B 2.17, which demonstrated an association between this particular CpG and the PAH studied in Meng et al. (2021), benzo[a]pyrene. This is due to benzo[a]pyrene being metabolically activated to benzo[a]pyrene diol epoxide (BPDE), forming covalent DNA adducts and binding plasma albumin, which was associated with altered methylation at CpG annotated to ACBD6. This link is explained by the metabolic activation of benzo[a]pyrene to benzo[a]pyrene diol epoxide (BPDE), which is a reactive metabolite that can form covalent DNA adducts and bind to the blood protein albumin. BPDE-albumin levels were associated with altered methylation at a CpG annotated to ACBD6 in Meng et al. These results were substantially evident in lung cancer patients due to the case-control nature of the study, which emphasizes the lung-cancer-related ACBD6 gene. The study could have also tested whether B[a]P-related methylation changes influenced lung-cancer-risk, because it included both lung cancer cases and controls.
Evidence on PAH exposure's association with DNA methylation is still limited due to different CpGs being analyzed separately, making results vary across studies. The loci with the most DNA methylation changes were AHRR, F2LR3, and ACBD6, genes often associated with tobacco and lung cancer. Smoking persisted as a significant source of heterogeneity because it independently affects methylation at multiple loci and contributes to PAH exposure.

Lung cancer-related genes associated with DNA methylation changes also have results that suggest an association with PAH exposure might be possible, even with slight DNAm changes. However, PAH specificity with possible DNA methylation changes was not reported other than for most PAHs other than 1-OHP, fluorene, and benzo[a]pyrene, since most studies focused on several PAHs or PAH-PM2.5-industrial-pollutants mixtures. Because hypomethylation and hypermethylation were both reported, with reduced methylation being more prevalent across studies, increased PAH exposure may be associated with these epigenetic changes.
Also, since many different CpGs were analyzed, the approach remained comprehensive, and although direct comparison and replication were challenging, this is an implication that studies are needed to corroborate the evidence on CpGs' possible relationship with PAH exposure. Moreover, ambiguous results found the occurrence of both increases and decreases of DNA methylation, meaning that replication is needed to ensure the accuracy of such results (Guo et al., 2022b). All in all, the evidence points toward an association between PAH exposure and DNA-methylation changes, but significant heterogeneity across tissues, exposure sources, CpG sites, and effect direction prevented the establishment of a single uniform PAH-induced methylation pattern.

V. Results II - Epigenetic Aging, PAH, and Lung Cancer

a. Polycyclic Aromatic Hydrocarbon Exposure and Epigenetic Aging

Biomarker-based and occupational studies generally support an association between high polycyclic aromatic hydrocarbon (PAH) exposure and accelerated DNA methylation aging. Li et al. (2018) provide evidence of PAH exposure assessed using urinary metabolites, which reflect internal dose. Higher concentrations of 1-hydroxypyrene and 9-hydroxyphenanthrene were independently associated with greater DNA methylation age acceleration. A one-unit increase in 1-hydroxypyrene was associated with approximately 0.53 additional years of AgeAcc (95% CI: 0.18-0.88). Campisi et al. (2023) reported a similar pattern in Polish male coke-oven workers with chronic occupational PAH exposure. Workers were more likely than controls to have a DNAmAge exceeding their chronological age (40.8% vs. 15.6% p= 0.007). Moreover, cumulative occupational exposure, calculated using urinary 1-pyrenol and years of coke-oven employment, was positively associated with DNAmAge. Together, these studies show consistent evidence of accelerated DNA methylation aging despite using different PAH exposure measures, strengthening the consistency of the association. However, the observational, cross-sectional design limits conclusions about temporality and causality. In addition, both studies examined relatively specific populations, including Chinese cohorts and male Polish coke-oven workers, so differences in ethnicity, sex, occupational setting, and exposure intensity may limit the generalizability of the findings for broader populations.

Furthermore, household combustion provides additional evidence for PAH-related GrimAge acceleration. Blechter et al. (2023) examined never-smoking women from Xuanwei, China, a region with substantial household air pollution from smoky-coal combustion. The researchers collected household fuel-use information by questionnaire and used validated exposure models to estimate concentrations of household air pollutants, including a cluster of 31 PAHs representing current exposure. Using leukocyte genomic DNA from blood samples of the participants, the study found that PAH exposure was most clearly associated with GrimAge acceleration. Specifically, a one-standard-deviation increase in the PAH cluster was associated with 0.77 years greater GrimAge acceleration (95% CI: 0.36-1.19). Importantly, the association was observed among never-smokers, suggesting that the contribution was independently due to household combustion-related sources. These findings are broadly consistent with studies using urinary PAH metabolites and occupational exposure methods, but the household study showed stronger and more consistent associations with GrimAge than with some epigenetic clocks based primarily on chronological age. GrimAge includes mortality and smoking-related methylation surrogates, which could make it more responsive to environmental exposures, suggesting that GrimAge is particularly sensitive to combustion-related exposure in these environments. Nevertheless, the study was based on a relatively small and geographically specific population of women from Xuanwei, which may limit the generalizability of the findings to other populations and exposure settings.

Continuing, Patil et al. (2025) provide evidence that the relationship between PAH exposure and epigenetic aging may also differ according to developmental timing. The study examined 43 children aged 36-60 months and assessed PAH exposure using urinary 2-naphthol and 1-hydroxypyrene.

Epigenetic aging was evaluated in buccal-cell DNA using several measures, including the Horvath, Skin&Blood, Pediatric Buccal Epigenetic (PedBE), and DNA methylation-based telomere length (DNAmTL) clocks. The strongest positive association was observed for current childhood exposure, with a two-fold increase in urinary 2-NAP associated with 0.21 years greater PC Skin&Blood epigenetic age deviation (95% CI: 0.03-0.39). In contrast, prenatal PAH exposure estimates generally suggested decelerated epigenetic aging, although these associations were not statistically significant. This difference in direction suggests that the effects of PAH exposure on epigenetic aging may depend on the developmental period in which exposure occurs. However, the small sample size, imprecision of prenatal estimates, and potential for residual confounding limit firm conclusions, making an apparent contrast between prenatal and current childhood exposure an important area for further research.

Finally, associations between PAH exposure and epigenetic aging are not fully consistent across different methylation clocks, suggesting that the observed effects may depend partly on the specific aging measure used. Shi et al. (2022) examined chemical exposure (PAHs, phthalates, nitroaromatics, and other contaminants) and epigenetic age acceleration among 73 healthy Chinese adults using several commonly applied methylation clocks, including the Hannum, Horvath, DNAmPhenoAge, and DNAmGrimAge measures. Although multiple clocks were assessed within the same population, only DNAmPhenoAge acceleration showed a significant positive association with chemical exposure. In the weighted quantile sum mixture analysis, an interquartile-range increase in the overall chemical-exposure index was associated with 1.31 years greater DNAmPhenoAge acceleration (95% CI: 0.02-2.60). Notably, the lack of significant associations for the Hannum, Horvath, and DNAmGrimAGe clocks within the same participants and exposure assessment suggests that heterogeneity across studies may reflect differences in the sensitivity of the epigenetic clocks themselves, rather than differences in population characteristics for exposure levels alone. This finding supports an association between airborne chemical-mixture exposure and accelerated epigenetic aging, but provides weaker evidence for PAHs specifically because PAHs were assessed alongside several other contaminants. The absence of consistent associations across all measures indicates that different clocks may not respond equally to the same environmental exposures. Taken together, the findings from Shi et al. suggest that conclusions about PAH-related biological aging may depend on the clock used to assess age acceleration. Overall, the evidence for PAH-related epigenetic aging is generally supportive but not fully consistent across aging measures. Among the studies evaluating multiple clocks, the clearest associations were observed for GrimAge and DNAmPhenoAge (Blechter et al. 2023, Shi et al. 2022), suggesting that health and mortality-related clocks may be particularly responsive to environmental exposures. However, the small number of observational studies and heterogeneity across clocks limit firm conclusions.

b. Epigenetic Aging and Lung Cancer Risk

Prospective observational studies generally support an association between accelerated epigenetic aging, particularly GrimAge acceleration, and subsequent lung cancer risk, including among never-smokers, although evidence for a direct causal relationship remains limited. Using DNAm phenotypic age (PhenoAge), DNAm GrimAge (GrimAge), and the mortality risk score (MRscore), Li et al. provide evidence for this pathway. Over 17 years of follow-up, investigators identified 146 incident lung cancer cases and observed positive dose-dependency across all three DNA methylation-based aging

measures. The strongest association was observed for the MRscore-8CpGs, in which each one-standard-deviation (SD) increase was associated with approximately a 106% higher hazard of lung cancer (HR = 2.06, 95% CI: 1.39-3.06). These associations persisted after adjustment for established cancer risk factors, including smoking. However, because the study is observational and epigenetic aging was measured in peripheral blood rather than lung tissue, the findings establish epigenetic age acceleration as a prospective biomarker of lung cancer susceptibility rather than a causal mediator of lung carcinogenesis.

Associations in never-smokers suggest the epigenetic age and lung cancer relationship is not explained entirely by self-reported smoking, although effects vary by clock. Rahman et al. (2024) provide evidence of this by studying the relationship solely using never-smokers as participants. The nested case-control study included 80 incident lung cancer cases and 83 controls from two prospective studies in Shanghai, with DNA methylation measured in prediagnostic oral-rinse samples. The researchers evaluated five epigenetic clocks, but the clearest association was observed for GrimAge. Each additional year of GrimAge acceleration was associated with 19% higher odds of lung cancer (95%CI: 1.06–1.34), and the association remained consistent after further adjustment for secondhand smoke exposure. Importantly, because all participants were never-smokers and the study didn't directly measure PAH exposure, this finding indicates that accelerated epigenetic aging is directly associated with lung cancer, even if not precipitated by PAH exposure. However, Bernstein et al. (2022) challenge this finding in a Mendelian randomization study. Using genome-wide association data from a meta-analysis, the study measured epigenetic age acceleration using HannumAge, Horvath Intrinsic Age, PhenoAge, and GrimAge on multiple cancers. Again, GrimAge provided the clearest outcomes, but it only provided evidence of associations to human cancers in general, not lung cancer specifically (GrimAge IVW OR = 1.00, 95% CI 0.91–1.09, p = 0.96; PhenoAge IVW OR = 0.97, 95% CI 0.94–1.00, p = 0.06; and HannumAge IVW OR = 0.99, 95% CI 0.95–1.04, p = 0.82). Findings for other clocks and cancers were inconsistent, meaning that further work would be required to explore the causes for these results. Importantly, this limits conclusions about causality between epigenetic aging and lung cancer, only supporting a moderate relationship between the two. Overall, result certainty is moderate for an association but low for a direct causal relationship, depending on the study analyzed.

VI. Results III - Polycyclic Aromatic Hydrocarbons and Lung Cancer

Across all the included studies, PAH exposure was usually linked to an increased lung-cancer risk or biomarkers indicative of increased risk of developing lung cancer. Evidence most directly linking PAH exposure to lung-cancer risk comes from studies in humans using diagnosed lung cancer as an outcome measure along with occupational, residential, or internal PAH exposure, or studies measuring tumor-specific biomarkers and/or PAHs in lung tissue. In these studies, PAH exposures were quite variable, but there was some consistency regarding the direction of evidence: increasing or higher PAH exposures were linked to higher lung-cancer risk, but depending on the smoking status of the study participants, the source of exposure, gender, and the dose of PAHs measured, the magnitude of association differed.

Occupational PAH evidence was best supported by the pooled SYNERGY study performed by Olsson et al. (2022) involving 16,901 lung cancer patients and 20,965 controls in 14 case-control studies

across Europe and Canada. Cumulative exposure to B[a]P served as a marker of occupational PAH exposure and was positively linked with lung-cancer risk after adjusting for cigarette smoking and other occupational lung carcinogens. The sensitivity analyses showed odds ratios (ORs) of 1.11 in males and 1.21 in females. Squamous-cell and small-cell lung cancers had strong associations with PAH exposure. Smoking has a notably greater increase in lung-cancer risk than solely occupational PAH exposure, but participants with both exposure types often had the highest ORs. The ORs in males were 9.3 and 10.59 for ever-smokers with or without occupational PAH exposures, respectively; for females, the corresponding ORs were 4.41 and 6.81. However, among never-smokers, the ORs of 1.1 and 0.9 were shown in males and females, respectively. Occupational research in this area has supported the above findings using biomarkers rather than actual diagnosis of cancer. According to Moubarz et al. (2023), there was a comparison of 147 secondary-aluminum workers with 30 administrative controls, and they had a serum BPDE-albumin adduct concentration of 19.5 ng/mL in the exposed workers compared to 10.52 ng/mL in the control workers ($p = 0.01$). In addition, higher BPDE-albumin concentration was associated with the cancer markers SCCAg and CCNB1 ($p < 0.0001$). Similarly, in the study by Alhamdow et al. (2019), differential expression of cancer markers was found in the serum proteins of non-smoking chimney sweeps occupationally exposed to PAHs. However, they are less direct than Olsson et al. since they used susceptibility and/or tumor-related biomarkers instead of lung-cancer occurrence.

Evidence from household exposure was especially important and independent of active smoking. In a study by Vermeulen et al. (2019), 1,015 never-smokers among women from Xuanwei and Fuyuan in China were exposed to household pollution with PAH due to burning smoky coal. A cluster of 25 PAHs showed lung cancer with an odds ratio of 2.21 in each standard deviation increase. For single pollutants, 5-methylchrysene exhibited the highest odds ratio (OR=5.42), whereas OR for benz[a]anthracene was 3.15 and OR for benzo[g,h,i]perylene was 1.84. The analysis accounted for age, secondhand smoke, and socio-economic status. As the population included never-smokers only, it was demonstrated that the lung cancer-PAH association could be found even without active smoking. At the same time, PAHs in exposure clusters were highly correlated; therefore, it was difficult to determine whether it was 5-methylchrysene or any other pollutant that caused the effect.

The internal PAH biomarker measurements gave another direct measure of PAH exposure, providing stronger evidence. Meng et al. (2021) quantified BPDE-albumin adducts, an internal biomarker of B[a]P activation, among two independent case-control groups of lung-cancer cases and controls. One standard deviation increase in BPDE-albumin adducts was significantly associated with lung cancer odds ratios of 1.57 and 1.35 in each set and 1.46 ($p=0.005$) in combination after adjusting for variables such as age, sex, BMI, smoking, and drinking. The study was highly relevant as exposure to PAHs and lung cancer disease had been measured among the same group of individuals. The methylation analysis also discovered five CpGs that mediated the association between B[a]P and lung cancer, though the pathway is discussed further in the DNA-methylation results section. A similar Chinese case-control study utilized BPDE-albumin adducts and found plasma microRNAs that statistically mediated a part of the association between B[a]P exposure and lung cancer risk (Xiao et al., 2024). These biomarker studies decrease the uncertainty due to the estimation of exposure based on occupations or questionnaires, though a limitation of case-control designs regarding temporality remains.

Environmental and target-organ evidence for this hazard was also available but more diverse. In a recent paper, a measurement of 16 PAHs accumulated in particulate matter found in human lung tissue samples (Zhang et al., 2023). The potential for causing cancer increased significantly with age; the benzo[a]pyrene equivalent potency of the PAH mix in people aged 70-80 was about 17-fold higher than that in people younger than 50. Thus, there is direct evidence showing accumulation of PAHs in the target organ, even though this is cross-sectional and limited to lung surgery patients, which makes it less useful for calculating prospective cancer risk. Prospective biomarker evidence was present in the form of HSA adducts altered due to oxidative stress in 573 never-smoking women from the Shanghai Women's Health Study who had lung cancer and whose plasma was collected 14 years before the diagnosis was made (Imani et al., 2025). The alterations in HSA adducts were associated with PAH biomarkers in urine. Since this exposure did not relate only to isolated PAHs but rather to urban air pollution and many co-exposures, this evidence should be considered environmental rather than PAH-specific.

On the whole, PAH to lung cancer pathway is one of the most well-supported pathways in the literature. Clinically, it was demonstrated by Olsson et al. (2022) in relation to occupational exposure, and Vermeulen et al. (2019) in relation to household exposure of non-smokers, as well as Meng et al. (2021), who found evidence for an internal exposure biomarker. Among the studies, the most important areas of heterogeneity included smoking status, age, sex, SES, occupational exposure to different substances, passive smoking, dietary patterns, intensity of exposure, and exposure to single and mixed PAHs. Irrespective of heterogeneity, positive results from occupational, household, internal exposure biomarker, and lung tissue studies have offered evidence on the connection between PAH exposure and increased risk of lung cancer. However, the evidence has not indicated causation as much as it showed association because case-control and cross-sectional studies were present in the studies, and PAH exposure usually went along with other carcinogens.

VII. Results IV: Mechanistic Evidence

a. PAH Metabolic Activation and Reactive Metabolite Formation

Relatively less reactive parent PAHs, such as benzo[a]pyrene, are metabolized into more reactive intermediates by enzymes in the cytochrome P450 (CYP) pathway after they enter the body. CYP1A1 is a gene regulated by the aryl hydrocarbon receptor (AhR) and encodes the cytochrome P450 1A1 enzyme, which contributes to the metabolism and activation of PAHs (Androutsopoulos et al., 2009). Cytochrome P450 1A1 converts less reactive pre-carcinogens from smoke or pollution, for instance PAHs, into reactive carcinogenic metabolites that can interact with cellular macromolecules and form adducts (Shimada and Fujii-Kuriyama, 2004). Benzo[a]pyrene, one of the most widely studied PAHs, can be metabolized into benzo[a]pyrene diol epoxide (BPDE), which is a highly reactive metabolite that can form DNA and protein adducts (Słowikowski et al., 2021).

Moreover, PAH exposure may change the expression and regulation of enzymes involved in the metabolism of reactive metabolites. Liu et al. (2026) reported evidence consistent with this claim, since

higher PAH exposure was associated with altered methylation in CYP1A1, GSTP1, and GSTM1, accompanied by mRNA hypoexpression. However, these genes play different roles in the process of metabolizing because CYP1A1 metabolically activates PAHs, while GSTP1 and GSTM1 detoxify reactive metabolites. This evidence indicates that PAH exposure may change the regulation of metabolic genes, altering the balance between PAH activation and detoxification, and increasing the burden of reactive metabolites on the body. Human biomarker evidence in Meng et al. (2021) demonstrates that benzo[a]pyrene is metabolically activated into reactive BPDE internally through the measurement of plasma BPDE-albumin adducts. The detection of BPDE-albumin adducts suggests that reactive metabolites are currently present and can covalently bind to cellular structures. Even though BPDE is capable of also forming DNA adducts, Meng et al. (2021) did not directly measure BPDE-DNA adducts or mutations, making the mechanistic connection between PAH metabolism and genotoxic mutations plausible yet not directly supported by evidence.

b. Oxidative Stress and DNA Damage

PAH exposure was repeatedly associated with oxidative stress and DNA-damage biomarkers across very different human exposure settings. The consistency of findings across exposure settings strengthens the biological plausibility and reliability of the evidence. For instance, Sambiagio et al. (2022) provide human biomonitoring evidence to further support this link by studying smokers using urinary 1-hydroxypyrene (1-OHP) as a biomarker for PAH exposure. The study found that higher urinary 1-OHP levels were associated with higher oxidative stress biomarkers. In fact, a 50% increase in 1-OHP corresponded to an 11-13% increase in oxidative stress biomarkers (Sambiagio et al. 2022). One of the most relevant biomarkers measured was 8-oxodG, which is formed when guanine is oxidized during ROS-DNA damage and is often utilized as a biomarker for oxidative DNA damage. This provides strong human biomonitoring evidence connecting greater internal PAH exposure and greater oxidative damage. Furthermore, Miranda-Guevara et al. (2023) reported similar oxidative DNA damage findings in a real-world population naturally exposed to coal dust. The study compared oxidative stress and DNA damage levels in residents living near a coal mine with controls. Exposed individuals exhibited greater oxidative stress, DNA damage, and shorter telomeres. All of these results are downstream evidence for PAH-exposure-induced cellular damage, especially since telomere shortening is an indication of oxidative stress accumulating over time. However, since coal dust is a mixture, these effects cannot be isolated to PAHs and might be attributed to other environmental pollutants. Likewise, Esteves at al. (2025) is another human study that demonstrated increased systemic DNA damage. However, the study was in a different exposure setting since it focused on wildfire-exposed firefighters who showed 22% higher comet assay DNA damage. Similar to Miranda-Guevara et al. (2023), Esteves et al (2025) measured a mixed exposure of wildfire smoke, where the corresponding effect cannot be solely attributed to PAHs nor can causation be inferred. Thus, the consistency across different settings strengthens the biological plausibility of oxidative stress and DNA damage as intermediate mechanisms. However, the mixed pollutant exposures in several studies limit the ability to prove causation for this pathway.

c. Direct Genotoxicity and Cell-Cycle Disruption

Reactive PAH metabolites can directly cause genotoxic injury by covalently binding to DNA regardless of ROS-induced DNA damage. This mechanism was demonstrated in Bauer et al. (2022) when human lung epithelial cells were experimentally exposed to B[a]P alone or with lower-molecular-weight PAHs. The study found that maximum anti-BPDE-DNA adduct formation was 674 ± 95 adducts per 10^8 nucleotides at 1.25 μM B[a]P in A549 cells. BPDE can create DNA lesions and adducts when it covalently binds to DNA, which can contribute to mutations and genomic instability if not properly repaired in time. Additionally, the researchers observed micronuclei formation in PAH-exposed cells, adding supplementary evidence of chromosomal and genomic damage. Bauer et al. also found an abnormally high proportion of cells in S phase, highlighting a disruption to the normal cell cycle during DNA replication after PAH exposure. Disruption to the cell cycle can increase the chance that damaged DNA is replicated which can contribute to the accumulation of mutations and genomic instability, and therefore adds another carcinogenic mechanism that PAH exposure can promote. This study provides direct experimental evidence connecting PAH exposure to DNA adduct formation, chromosomal damage, disruptions in DNA replication, and strengthens the biological plausibility of PAH-induced carcinogenesis. However, the study was done in vitro; it only shows the biological mechanism under controlled laboratory conditions. Consequently, the study does not indicate that the same effects could be reproduced at environmentally realistic doses in humans nor that it could directly cause lung cancer.

d. Epigenetic Alterations

The extent of PAH-induced damage may extend past just direct cellular and DNA damage and into epigenetic regulation. Alhamdow et al. reported that occupational PAH exposure was associated with the hypomethylation of AHRR and F2RL3. AHRR is a part of the aryl hydrocarbon response system, so changes in the regulation of this gene affect how cells respond to PAHs. Additionally, Liu et al. (2026) demonstrated that PAH-induced methylation changes in CYP1A1, GSTP1, and GSTM1 were associated with mRNA hypoexpression. Moreover, epigenetic changes were also found in mitochondrial DNA in Guo et al. (2022), where PAH exposure was associated with altered mtDNA methylation, especially MT-CO1-related methylation changes at the MT-CO1-P2 locus. This finding suggests that PAHs' effects on epigenetics may extend beyond nuclear DNA to mtDNA regulation, which regulates oxidative phosphorylation and cell energy metabolism. These findings indicate that PAH-induced epigenetic changes can happen in both nuclear and mitochondrial loci, especially genes involved in mitochondrial function and metabolism. Together, these findings suggest that PAH-associated epigenetic alterations occur across both nuclear and mitochondrial loci, including genes involved in xenobiotic metabolism and mitochondrial function. However, the methylation changes were heterogeneous in terms of tissues, loci, exposure settings, effect direction, and both hypermethylation and hypomethylation were present.

e. Broader Epigenetic and Cellular Effects

Finally, PAH exposure was experimentally shown to include epigenetic changes associated with a malignant phenotype in Xu et al (2025). The study found that chronic B[a]P exposure was linked to a RNA m6A methylation, which promotes the malignant growth of bronchial epithelial cells through increased proliferation, migration, and invasion. This evidence supports a broader epigenetic pathway than a specific DNA methylation aging pathway, because m6A is an RNA modification. Xu et al. (2025) serves as strong mechanistic evidence for the epigenetic pathway that B[a]P can affect epigenetic regulation linked to carcinogenic cellular phenotypes, because it extends the mechanism further than more human observational studies. Overall, the mechanistic evidence supports many interacting pathways instead of one single route, beginning with reactive metabolite formation, ROS-induced DNA damage, formation of DNA adducts, both nuclear and mitochondrial DNA epigenetic changes, regulation of RNA m6A, disruptions to genomic stability, and changes to the cell cycle.

VIII. Results V: Integrated Pathway

a. Strength of Mechanistic Evidence

One of the strongest mechanistic associations was between reactive PAH metabolites and oxidative stress or DNA damage, because human monitoring, real-world exposure, and experimental evidence all supported the link. While one study provided more PAH-biomarker evidence for oxidative stress (Sambiagio et al., 2022), other studies reported DNA-damage effects in different real-world exposure settings using an exposure mixture instead of measuring isolated PAHs (Miranda-Guevara et al., 2023; Esteves et al., 2025). Additionally, Bauer et al. (2022) strengthen this notion experimentally by showing how controlled B[a]P exposure can directly cause DNA adducts, micronuclei, and disruptions to the cell cycle. The association between PAH exposure and epigenetic changes is moderately supported and mechanistically plausible because multiple studies show PAH-associated epigenetic changes. However, fewer studies directly establish oxidative stress as the intermediate mechanism linking PAH exposure to those exact epigenetic alterations. Overall, the association between PAH and DNA methylation changes is more strongly supported than the link between reactive oxygen species triggered by PAH exposure and DNA methylation changes.

b. PAH Exposure and Epigenetic Age Acceleration

Numerous studies in our evidence base have reported the association between higher PAH exposure and accelerated epigenetic aging, making the pathway moderately supported (Li et al. 2018; Shi et al. 2022; Blechter et al. 2023; Campisi et al. 2023; Ding et al. 2026). The studies utilized various PAH exposure measures, populations, and epigenetic aging clocks. However, the overall direction of association was comparable enough to support the association between PAHs and epigenetic aging despite methodological differences. Yet, these studies generally measured PAH exposure and epigenetic aging, but they did not directly test whether oxidative stress, DNA damage, or certain PAH-induced methylation changes caused the epigenetic age acceleration. Therefore, epigenetic aging should be considered a potential downstream effect of accumulated or a correlate of PAH-associated epigenetic dysregulation rather than being a causal intermediate mechanism in PAH-induced lung carcinogenesis.

c. Epigenetic Aging and Lung-Cancer Risk

The final step of the potential pathway is whether epigenetic age acceleration is associated with later lung risk. In a study, aging-related methylation measures were positively correlated with incident lung cancer (Li et al. 2022). Other research adds supporting evidence in never smokers which suggest the epigenetic aging and lung cancer link is not only caused by active smoking (Rahman et al., 2024). However, Mendelian-randomization findings did not support a causal effect of epigenetic age acceleration on lung cancer, so epigenetic aging is better supported as a risk-predictive biomarker rather than a proven cause.

d. Overall Pathway Certainty

The strongest evidence supports the association between PAH and oxidative stress and DNA damage, epigenetic alteration and lung-cancer risk, however evidence modestly supports the associations between PAH exposure and epigenetic age acceleration and between epigenetic age acceleration and lung cancer. Finally, the pathway for oxidative DNA damage to certain methylation alteration causing epigenetic age acceleration and lung cancer is the most indirect and inferred. All in all, the complete pathway is biologically plausible but is still inferential because no included studies measured all major intermediates in one chain nor do they provide the full sequence in one cohort.

IX. Discussion

Across the included evidence, the most consistent finding was the association between PAH exposure and epigenetic alterations, while evidence supporting the complete pathway from exposure to lung cancer is indirectly supported. Through synthesis of complementary studies, there is evidence supporting biological plausibility to connect PAH exposure with carcinogenic risk.

Multiple studies with varied models support the mechanistic link of the pathway. Sambiagio et al. (2022) linked PAH-related tobacco exposure with oxidative-stress biomarkers and ROS-related damage, and Miranda-Guevara et al. (2023) associated coal-dust exposure with oxidative stress, DNA damage, and shortened telomeres.

The evidence extends toward epigenetic alteration as well. There were associations with internal PAH, nitrated-PAH, and oxygenated-PAH exposure with mitochondrial DNA (mtDNA) methylation in young adults, supporting consistently that oxidative stress and genomic damage occur as biological responses associated with exposure (Guo et al., 2022). However, there is less direct support that genomic changes may translate into epigenetic changes. While the transition from acute genomic damage to sustained epigenetic alterations requires further validation, the recurrent identification of methylation changes in F2RL3/AHRR, CYP1A1/GSTP1/GSTM1, and MT-ATP6/8 across independent studies is highly critical. This recurrence shows across study populations and strengthens the association between PAH exposure and disrupted metabolism, increased oxidative stress, and mitochondrial dysfunction.

Across the aging studies, PAH exposure was generally associated with accelerated epigenetic aging across different populations and study designs, including studies using DNAmAge, DNAmPhenoAge, Horvath, inter alia. (Shi et al. 2022, Patil et al. 2025, Blechter et al. 2023, Campisi et al. 2023, Ding et al. 2026, Khan et al. 2026), with slight variation depending on the study and what the

clock measured (biological age, health, etc.). That said, the epigenetic age-to-cancer pathway is less certain. Prospective observational studies associated accelerated epigenetic aging with subsequent lung cancer (Rahman et al. 2024, Li et al. 2022), while the Mendelian-randomization study did not support a direct causal effect, where HorvathAge predicted a decreased risk of lung cancer, and other clocks did not find evidence of causality. (Morales-Bernstein et al. 2022).

The methylation evidence may provide an association between PAH exposure and downstream lung-cancer risk. Five CpGs were identified which statistically accounted for ~58% of the association between benzo[a]pyrene and lung cancer (Meng et al., 2021). While this statistical mediation analysis is consistent with a potential site-specific methylation-mediated pathway, it originates from observational data. It does not establish biological causality, nor does it imply that generalized epigenetic age acceleration serves as the functional mediator. Similarly, a link between F2RL3/AHRR methylation, a strong predictor of lung cancer, with occupational PAH exposure was present (Alhamdow et al., 2020). However, this study inferred lung cancer risk through validated surrogate DNA methylation markers rather than directly measuring clinical cancer incidence, making the evidence for downstream lung cancer indirect. In contrast to studies relying on intermediate biomarkers, epidemiological investigations directly measuring clinical endpoints confirm a positive association between PAH exposure and lung cancer incidence across both occupational and household environments. However, biological plausibility cannot be confused with proof of the complete mechanism.

Synthesizing these complementary lines of evidence yields a biological model where PAH-induced oxidative stress drives epigenetic modification. The strength of this model stems from the repeated convergence of distinct studies on identical gene networks, even if the step-by-step temporal progression remains sparsely explored in human cohorts.

The downstream evidence suggests that these alterations may be relevant to lung carcinogenesis, but the complete sequence should not be interpreted as proven causation. The strongest link in the pathway is PAH-to-oxidative stress and DNA damage/lung cancer, while there is a moderate link between PAH-to-epigenetic aging, with the weakest aspect being when aging is the mediator linking PAH exposure to lung cancer.

Ultimately, pathway certainty decreases as it advances downstream. While early biological responses such as internal PAH exposure leading to mtDNA methylation (Guo et al., 2022) and oxidative stress (Zhou et al. 2026; Miranda-Guevara et al. 2023) are strongly supported, connecting these early responses to clinical lung cancer relies on independent epidemiological studies (Vermeulen et al. 2019; Olsson et al. 2022) that did not track intermediate epigenetic markers.

For the PAH exposure-to-epigenetic aging-to-lung cancer portion of the pathway, certainty is more limited due to many studies containing a cross-sectional design, varied exposure sources and measurements of exposure, different clocks, and other pollutants/exposure sources. The aging evidence itself was considered strong for an association between PAH exposure and epigenetic age acceleration, but the downstream causal relationship is weaker because of varied conclusions in prospective observational studies supporting a positive association, whereas the Mendelian-randomization study weakens evidence for causation.

X. Research Gaps

Though the PAH exposure-to-oxidative stress-to-epigenetic change-to-carcinogenic risk pathway has been explored by various studies, the complete pathway is generally not measured within a single study. The proposed pathway is generally reconstructed across complementary studies rather than tested end-to-end, making the connections between individual pathway steps dependent on synthesis across studies. Additionally, most of the mechanistic evidence is cross-sectional or observational, making it difficult to establish a clear causal sequence from PAH exposure to oxidative stress/DNA damage to epigenetic alteration and ultimately to carcinogenesis.

A significant methodological gap is the reliance on single, spot-urinary PAH metabolite measurements. Because urinary biomarkers reflect acute exposure over hours or days, single-time-point measurements fail to capture the chronic, cumulative exposure required to drive long-latency outcomes such as epigenetic drift and carcinogenesis. These studies measured different outcomes, including oxidative-stress biomarkers, DNA damage, mtDNA copy number, mtDNA methylation, RNA methylation, miRNA expression, and tumor-related biomarkers.

There is a significant deficit in the amount of studies exploring epigenetic age acceleration as a mediator in the PAH exposure-to-lung cancer pathway, serving as the principal gap. In addition, human epigenetic studies use peripheral blood rather than lung tissue, creating uncertainty on how well aging changes are reflected on the target organ. There is also a lack of repeated exposure measurement, with most studies relying on a singular measurement from a recent exposure event.

The variations in PAH-induced epigenetic aging reflect differences in study design. Clock choice dictates whether the metric captures chronological drift or phenotypic decline, exposure setting dictates acute toxicity versus chronicity, and tissue selection (blood or lung) determines whether the signal reflects inflammation or target-organ pathology.

Articles also reported a different direction for prenatal exposure, with decelerated rather than accelerated epigenetic aging (Patil et al., 2025). Timing of exposure may explain the opposite direction and may affect developmental epigenetic trajectories differently from occupational and environmental exposure. There is limited evidence on whether methylation changes persist or reverse after exposure cessation (Esteves et al. 2025). This suggests that the timing of exposure plays a crucial role, where early-life exposure during sensitive developmental windows disrupts epigenetic trajectories differently than chronic exposure in adulthood.

Evaluating PAH-specific effects in human cohorts is further complicated by pervasive co-exposure confounding. Environmental and occupational PAHs routinely co-occur with tobacco smoke, fine particulate matter, coal dust, and industrial solvents, making it difficult to isolate the independent epigenetic impact of PAHs from multi-pollutant mixtures.

The transition from acute genomic damage to sustained epigenetic alterations remains unclear as well, with less direct evidence proving that DNA damage itself is what causes methylation changes.

XI. Future Directions

Future studies should measure multiple stages of the pathway within the same participants, ideally within prospective longitudinal cohorts. These studies should include repeated PAH exposures, oxidative stress, DNA damage, mitochondrial alterations, epigenetic changes, and relevant lung outcomes to strengthen causal inference.

Tracking exposure, intermediate oxidative stress, epigenetic alterations, and eventual clinical outcomes within the same cohort is essential to establish true temporal ordering and overcome the limitations of current cross-sectional data.

Greater consistency in biomarker selection, biological endpoints, and standardization of a core set of epigenetic clocks that each measure one aspect of aging changes (simple time passed since birth to physiological decline, systemic inflammation, and the pace of ongoing systemic deterioration) would improve cross-study comparison as well. Future methodologies should explicitly delineate distinct biological endpoints—categorizing acute DNA damage, mitochondrial modifications (mtDNA copy number and methylation), and nuclear epigenetic alterations as discrete mechanistic steps rather than treating them as interchangeable surrogates for biological response.

Alongside this, studies should examine both short-term and cumulative exposure. The shared pathway analysis suggests that stable epigenetic markers may not correspond closely with short-term urinary biomarkers, so incorporating both immediate and cumulative exposure measures could clarify how exposure duration influences the observed biological response. Future research must transition from single-pollutant analyses to multipollutant mixture models. This variety is essential to overcome PAH-specific attribution problems and isolate the independent biological effects of PAHs. Additionally, pairing these epidemiological advances with paired blood-and-lung tissue sampling will help confirm whether systemic signals accurately reflect target-organ pathology.

Finally, research should investigate whether PAH-associated methylation and aging changes persist or reverse after exposure cessation, and whether these changes independently predict later lung cancer. To avoid overstating the evidence, the proposed pathway must be evaluated across three distinct levels of certainty. Firstly, evidence directly linking PAH exposure to lung cancer incidence is strong and well-established. To follow, evidence connecting PAHs to site-specific methylation and epigenetic age acceleration is highly supportive, though heterogeneous. Lastly, evidence that epigenetic aging acts as a functional mediator of PAH-induced lung cancer remains the least established.

In summary, advancing this pathway requires moving beyond cross-sectional, single-biomarker studies. The main priority for future research is establishing prospective longitudinal cohorts that measure the whole pathway. In addition to that, studies must integrate repeated exposure measurements. Another priority would be paired blood-and-lung tissue sampling, closely followed by diversified populations across varied regions, exposure settings, genders, exposure levels, and long-term lung cancer follow-up.

XII. Conclusion

In summary, our systematic review aimed to address the correlation of human exposure to polycyclic aromatic hydrocarbons to epigenetic aging acceleration, DNA methylation, and lung cancer risks as our complete pathway has not been researched or comprehensively evaluated from end to end. Across the 33 reviewed studies, the overall evidence on PAH exposure is generally associated with epigenetic alterations and independently with lung cancer. However, these two have been determined through markedly different types of samples. This pathway is supported segment by segment, although not directly demonstrated in completion. Multiple studies within our list found a clear relationship between PAH exposure and altered DNA methylation, which is evident throughout multiple exposure

and methylation types across our reviewed works. The segmented pathways connecting PAH exposure to epigenetic aging, and PAH exposure to lung cancer, were both well supported individually, whereas evidence for the complete pathway remained with greater limitations. Methylation may partially mediate part of PAH-related effects across studies where it carries measurable weight on associations between the chemicals and their effects. There also exist various differences and patterns amongst studies: 14 of 33 studies were conducted in China along with occupational cohorts overrepresenting PAH exposure in general, which develops a population that is less generalized at the global scale. In addition, finding studies that individualized PAHs and their effects was difficult and prevented results from detecting more impacts surrounding each of these chemicals. Furthermore, 14 out of 33 studies include cross-sectional designs which leaves temporality less targeted throughout our review, hindering the normality of our study. With our research, a priority is availability to a prospective cohort which has measurable and environmental PAH exposure biomarkers, multiple epigenetic clocks, and lung cancer risks from the same participants. Such a design would permit direct testing of whether epigenetic age acceleration mediates the association between PAH exposure and lung cancer, remaining the central unexperimented question in this field.

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